



North South University

Electrical and Computer Science and Engineering

Course Name: CSE215

Faculty Name: Dr. Mohammad Shifat-E-Rabbi [MSRb]

Department Name: Electrical and Computer Science and Engineering

Group Number: 5

Group Name: Astro

Group Members:

Name	ID
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Project Repository: <https://github.com/touhidhimuiux/Student-Management-System-in-Java/tree/master>

1. Project Description

Our project, "Student Management System," is a comprehensive application designed to streamline and manage various aspects of student administration. The primary goal is to facilitate efficient handling of student-related information, course enrollments, and academic performance tracking.

Upon accessing the system, users are presented with a login page that ensures only authenticated users can access the system. After successful login, users enter the Student Management System home page, which features several key panels:

1. **Student JFrame:** This panel is used to manage detailed student information. It includes the operations to add, update search and delete student records. A table displays a list of all the students with their unique ID, name, department, gender and other personal information. This tab also allows the user to browse files and add a student image.
2. **Enroll JFrame:** This frame lets the user select and enroll a student in multiple courses for the semester. The frame displays a table of the current available courses and a table that shows the selected courses. It prevents duplicate entries of the same course and stops the user from exceeding the credit limit per semester. Additionally, they have the option to delete or retake a course from the selected ones if needed.
3. **Marks JFrame:** This panel is used to manage student grades and gpa. The tables display a list of all students and another table that displays the selected student's enrolled courses. The user can then assign a mark to each course. This mark is used to calculate their GPA and displayed in the student's profile. Users can also drop a course in this panel.
4. **Course JFrame:** This panel is used to manage course information. Users can search for and view available courses. It also has the option to add, update and delete courses. A table displays all the courses showing course code, name and credits. Selecting a course will display the list of students enrolled in that course with their academic details.
5. **Records JFrame:** This panel displays the list of all the students in the system and their academic information.

Overall, the Student Management System aims to enhance the efficiency and accuracy of managing student information, course enrollments, and academic performance, making it an indispensable tool for educational institutions.

3.Group Members' Contribution

Group Member Name	Contribution
Mahjabin Noor Nabila	Implemented the backend of the code. Added all functionalities in Student JFrame, Enroll JFrame, Course JFrame and Records JFrame and connected it to the Home Class Designed and implemented the following classes: Student, Student Manager Class, Course, CourseManager and Gradable.
Touhid Ara Himu	Designed the project's full GUI, developed the login and authentication system, connected it with the Home JFrame.

4. Drawbacks and Future Implementation Plan

Drawbacks:

1. **User Authentication check:** The system may not store user information.
2. **User Interface:** The current design is basic and not very user-friendly.
3. **Limited Course Management:** The system only allows basic course enrollment and deletion, lacking features like course prerequisites and notifications.
4. **Security:** The login system works but lacks advanced security features like two-factor authentication and data encryption.

Future Implementation Plan:

1. **Improving User Authentication:** We plan to optimize the database and use cloud services to handle more users efficiently.
2. **Enhancing User Interface:** We will redesign the interface to make it more modern and user-friendly, ensuring it works well on all devices.
3. **Advanced Course Management:** Future updates will include features like waitlists, course prerequisites, and notifications for course changes.
4. **Better Security:** We will add two-factor authentication, encrypt sensitive data, and improve logging and monitoring for better security.
5. **System Integration:** We aim to integrate the system with other educational tools to provide a more comprehensive experience.

These improvements will make the Student Management System more robust, user-friendly, and secure, meeting the needs of a larger user base.

UML Diagram:

